MIL-C-82644(OS)
27 January 1975
SUPERSEDING
NOTS XS 142 Rev E
1 April 1969

MILITARY SPECIFICATION

COMPOUNDS, EPOXY, ENCAPSULATING

This Specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers cured compounds based upon epoxy resins used for casting shapes, for encapsulating: (a) electronic parts and assemblies, (b) tranformers, (c) conductors, (d) coils, and for sealing terminal wires and connector contacts.

1.2 Classification. Compounds shall be of the following types, as specified (see 6.1):

Type I - General purpose
Type II - Low expansion
Type III - Hermetic sealing
Type IV - Free machining

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, publications, or such portions thereof as may be designated herein, of the issue in effect on the date of invitation for bids, form a part of this specification. In the event conflict between documents occurs, the requirements of the contract shall have precedence over this specification and this specification shall have precedence over any other document referenced herein.

SPECIFICATIONS

FEDERAL

VV-I-530	Insulating Oil, Electrical (for Transformers, Switches, and Circuit Breakers)

STANDARDS

FEDERAL

FED-STD-406	Plastics: Methods of Testing

FSC 8030

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MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

AMERICAN TRUCKING ASSOCIATIONS, INC., AGENT: CLASSES AND RULES

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification 11; Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 First article testing. Unless otherwise specified in the contract or purchase order, a sample of production material shall be submitted for first article testing.

3.2 Material. The material furnished shall consist of an epoxy-based polymer and a curing agent. Each may contain fillers or other ingredients necessary to yield compounds of the required properties.

3.3 Physical and electrical requirements. Samples cured according to manufacturer's instructions (see 3.5) shall conform to the requirements of Table I when subjected to the tests of 4.5.

3.4 Suitability. Compounds shall be suitable for application and use on electronic assemblies and individual parts. Compounds shall preserve the electrical properties of the equipment to which they are applied by sealing against such environmental conditions as moisture, dirt, fumes, fungus, atmospheric pressure changes, or combinations of these which may be encountered in the military services. Compounds shall be capable of mechanically supporting the part or assembly encapsulated and shall not require a case or other external support after curing has been completed. Compounds shall not cause deterioration of materials used in electronic assemblies and parts.

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Table 1. Physical and Electrical Requirements of Epoxy Compounds

PROPERTY	REQUIREMENT				TEST METHOD
	TYPE I	TYPE II	TYPE III	TYPE IV	
Flexural strength, psi, minimum	8,000	11,000	8,000	6,000	4.5.2
Izod impact strength, ft-lb/in of notch, minimum	0.30	0.35	0.30	0.25	4.5.3
Compressive strength, psi, minimum	15,000	25,000	18,000	20,000	4.5.4
Insulation Resistance					
73.5°F, ohms, minimum	10^{12}	10^{12}	10^{12}	10^{12}	4.5.5.1
200° F, ohms, minimum	10^{10}		10^{10}		4.5.5.2
250° F, ohms, minimum	10^{9}	10^{12}	10^{9}	10^{12}	4.5.5.3
300° F, ohms, minimum		10^{11}		10^{10}	4.5.5.4
Insulation resistance, specimen conditioned for 96 hours at 90 percent relative humidity and 95° F, ohms, minimum	10^{12}	10^{12}	10^{12}	10^{12}	4.5.5.5
Volume resistivity, ohm-cm, minimum	10^{14}	10^{14}	10^{14}	10^{14}	4.5.6
Dielectric strength, volts/mil, minimum	400	450	450	400	4.5.7
Dielectric constant, maximum					4.5.8
60 Hz	5.5	5.5	5.5	6.0	
1 kHz	5.5	5.5	5.5	6.0	
100 kHz	5.5	5.5	5.5	6.0	
Dissipation factor, maximum					4.5.8
1 kHz	0.04	0.04	0.04	0.04	
100 kHz	0.04	0.04	0.04	0.04	
Arc resistance, seconds, minimum	120	180	150	100	4.5.9
Heat distortion temperature, degrees fahrenheit, minimum	160	220	160	215	4.5.10
Coefficient of linear thermal expansion, length/unit length/°C, maximum					4.5.11
-53.9°C to 23°C	40×10^{-6}	40×10^{-6}	30×10^{-6}	50×10^{-6}	
23°C to 73.9°C	60×10^{-6}	45×10^{-6}	60×10^{-6}	55×10^{-6}	
Water absorption 24 hour immersion, percent maximum	0.10	0.08	0.08	0.10	4.5 12

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3.5 Manufacturer's instructions. When specified in the contract or order (see 6.2.2), the manufacturer shall furnish instructions for the casting of specimens which shall include:

a. Proportions of each ingredient

b. Sequence and temperature for mixing ingredients

c. Stirring required in mixing

d. Evacuating or centrifuging required to eliminate entrapment of air

e. Shelf life of mixed ingredients

f. Pot life of mixed ingredients

g. Pouring temperature

h. Curing time and temperature for casting each type of specimen

i. Mold release agents, mold agents, and mold materials required

j. Curing agent to be used

3.6 Workmanship. The workmanship shall be in accordance with high grade commercial practice covering this type of material. The ingredients shall be uniformly processed to produce the quality of product established by this specification. The finished product shall be clean and uniform, and free of any defects which might impair its utility such as sediment, or other foreign particles.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The tests required to ensure that the compound or lots of compound meet the requirements of this specification are classified as follows:

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a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Lot. The term "lot" shall mean "inspection lot"; that is, a collection of product submitted by a supplier for inspection. Unless otherwise specified in the contract or order, a lot shall consist of all material produced from a single batch or production run, and offered for delivery at one time.

4.3.2 First article sample. A first article sample shall consist of a sufficient quantity of compound ingredients to produce 200 cubic inches of cast specimens. Further production of the compounds, prior to approval of the first article sample by the procuring activity, shall be at the suppliers risk.

4.3.3 Testing. The first article sample shall be submitted to all the tests of 4.5 for conformance to all the requirements of Table I. When specified in the contract or order (see 6.2.2), a test report covering the first article inspection shall be submitted to the procuring activity.

4.3.4 Subsequent first article testing. First article testing shall be performed wherever a change is made in the manufacturing of starting materials or process, in this specification or as requested by the procuring agency.

4.4 Quality conformance inspection.

4.4.1 Quality conformance sample. A quality conformance sample shall consist of a sufficient quantity of compound ingredients to produce 100 cubic inches of cast specimens.

4.4.2 Quality conformance tests. The following quality conformance tests shall be conducted on samples provided in accordance with 4.4.1:

Test	Test Procedures
Flexural strength	4.5.2
Izod impact strength	4.5.3
Dielectric constant	4.5.8
Dissipation factor	4.5.8
Deflection temperature under load	4.5.10
Water absorption	4.5.12

When specified in the contract or order (see 6.2.2) a test report covering the quality conformance inspection shall be submitted to the procuring activity.

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4.4.3 Rejection. If test results obtained on the samples representing a lot do not conform with the requirements of Table I, this shall be cause for rejection of the lot.

4.5 Test methods.

4.5.1 Test conditions. Unless otherwise specified, all tests shall be conducted under standard laboratory conditions with relative humidity at 50 percent plus or minus 4 percent and a temperature of 73.5 degrees plus or minus 2 degrees Fahrenheit (23 degrees plus or minus 1.1 degrees Celsius).

4.5.2 Flexural strength. Test method 1031 of FED-STD-406 shall be used in measuring five specimens 0.25-inch by 0.5-inch by 5 inches. The average of five flexural strength measurements shall be reported (see 4.3.3 and 4.4.2).

4.5.3 Izod impact strength. Test method 1071 of FED-STD-406 shall be used in measuring five specimens, each having a depth of 0.500-inch plus 0.006-inch minus 0.000-inch, a width of 0.5-inch, and a length of 2.5 inches. The specimens shall be notched as indicated. The average of five izod impact strength measurements shall be reported. (see 4.3.3 and 4.4.2)

4.5.4 Compressive strength. Test method 1021 of FED-STD-406 shall be used in measuring five specimens 0.5-inch by 0.5-inch by 1 inch. The average of five compressive strength measurements shall be reported (see 4.3.3).

4.5.5 Insulation resistance. Test method 4041 of FED-STD-406 shall be used in measuring three specimens 0.125-inch thick and 4 inches in diameter. The electrodes shall be spaced 1.25-inches apart in the form of a triangle at the center of the disk so that three measurements can be taken on each specimen. The average of nine insulation resistance measurements shall be reported for each of the following conditions.

4.5.5.1 As cast after standard laboratory conditioning for 48 hours plus or minus 1 hour (see 4.5.1)

4.5.5.2 Conditioned for 30 minutes plus or minus 2 minutes at 200 degrees plus or minus 2 degrees Fahrenheit (93.3 degrees plus or minus 1.1 degrees Celsius) and relative humidity of 50 percent plus or minus 4 percent

4.5.5.3 Conditioned for 30 minutes plus or minus 2 minutes at 250 degrees plus or minus 2 degrees Fahrenheit (121.1 degrees plus or minus 1.1 degrees Celsius) and relative humidity of 50 percent plus or minus 4 percent

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4.5.5.4 Conditioned for 30 minutes plus or minus 2 minutes at 300 degrees plus or minus 2 degrees Fahrenheit (148.9 degrees plus or minus 1.1 degrees Celsius) and relative humidity of 50 percent plus or minus 4 percent.

4.5.6 Volume resistivity. Test method 4041 of FED-STD-406 shall be used in measuring five speciments 0.125-inch by 2 inches by 2 inches. The average of five volume resistivity measurements shall be reported (see 4.3.3).

4.5.7 Dielectric strength. Test method 4031 of FED-STD-406 shall be used in measuring five speciments 0.125-inch thick and 4 inches in diameter. The tests shall be made on specimens immersed in transformer oil conforming to VV-I-530. The short-time test to breakdown using 500 volts per second shall be conducted. Dielectric strength shall be reported as the minimum, maximum, and average of the five measurements in volts per mil. (see 4.3.3)

4.5.8 Dielectric constant and dissipation factor. The test circuit specified in test method 4021 of FED-STD-406 shall be used in measuring three specimens 0.125-inch by 5 inches in diameter. The dielectric constant shall be measured for each specimen at 60 hertz, 1 kilohertz, and 100 kilohertz. The average of the three measurements at each frequency shall be reported. The dissipation factor shall be measured at 1 kilohertz and 100 kilohertz for each of the three specimens. The average of the three measurements at each frequency shall be reported (see 4.3.3 and 4.4.2).

4.5.9 Arc resistance. Test method 4011 of FED-STD-406 shall be used in testing three specimens 0.125-inch plus or minus 0.010-inch thick and 4 inches in diameter. Three arc resistance readings shall be taken on each specimen and the average of nine readings reported (see 4.3.3).

4.5.10 Deflection temperature under load. Test method 2011 of FED-STD-406 shall be used in testing three specimens 0.5-inch by 0.5-inch by 5 inches. The test shall be run with a maximum fiber stress of 264 plus or minus 6 psi on the specimen. The average of three deflection temperature measurements shall be reported (see 4.3.3 and 4.4.2).

4.5.11 Coefficient of linear-thermal-expansion. Test method 2031 of FED-STD-406 shall be used in testing three specimens 0.5-inch by 0.5-inch by 2 inches. The coefficient shall be determined for the following temperature ranges:

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a. Minus 65 plus or minus 2 to plus 73.4 plus or minus 2 degrees Fahrenheit (minus 53.9 plus or minus 1.1 to plus 23 plus or minus 1.1 degrees Celsius)

b. Minus 9.4 plus or minus 2 to plus 165 plus or minus 2 degrees Fahrenheit (minus 23 plus or minus 1.1 to plus 73.9 plus or minus 1.1 degrees Celsius)

The average of the three coefficient measurements for each temperature range shall be reported (see 4.3.3).

4.5.12 Water absorption. Test method 7031 of FED-STD-406 shall be used to test three specimens 0.125-inch thick by 1 inch by 3 inches. The average of three measurements shall be reported (see 4.3.3 and 4.4.2).

4.5.13 Visual examination. Each of the samples taken in accordance with 4.3 shall be examined for conformance to the requirements of good workmanship specified in 3.6. Nonconformance is cause for rejection of the entire lot.

4.5.14 Examination of preparation delivery. A sample package from each lot shall be examined for compliance with Section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level C. Packaging shall be in accordance with the supplier's commercial practice when such practice meets the requirements of this level. Containers shall be in accordance with the National Motor Freight Classification, Uniform Freight Classification 11; Ratings, Rules and Regulations, or other carrier regulations as applicable to the mode of transportation.

5.2 Packing.

5.2.1 Level C. The subject commodity packaged as specified herein shall be packed in accordance with the National Motor Freight Classification, Uniform Freight Classification 11; Ratings, Rules and Regulations, or other carrier regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such practice meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order, the interior and exterior packages and containers shall be marked in accordance with MIL-STD-129. The marking shall include the following:

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a. Compounds, epoxy, encapsulating (Type_____)
b. Specification number and revision letter
c. Supplier's name
d. Supplier's code designation
e. Lot designation
f. Formulation date and expiration date
g. Storage instructions
h. Special handling precautions

6. NOTES

6.1 Intended use.

6.1.1 Type I. Type I compound is intended for general purpose applications where resistance to humidity is required. The mechanical and electrical properties are suitable for operating temperatures up to 200 degrees Fahrenheit (93.3 degrees Celsius).

6.1.2 Type II. Type II compound is intended for applications requiring minimum shrinkage and thermal expansion. The electrical properties and thermal shock resistance are suitable for operating temperatures up to 300 degrees Fahrenheit (148.9 degrees Celsius).

6.1.3 Type III. Type III compound is intended for hermetic sealing applications where adhesion and wetting characteristics are important. The electrical properties and thermal shock resistance are suitable for operating temperatures up to 200 degrees Fahrenheit (93.3 degrees Celsius).

6.1.4 Type IV. Type IV compound is intended for applications where machinability and heat resistance in the finished part are required. The mechanical and electrical properties are suitable for operating temperatures up to 300 degrees Fahrenheit (148.9 degrees Celsius).

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

a. Title, number and data of this specification.
b. Type required (see 1.2)
c. First article sample if required (see 3.1)

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraphs:

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Paragraph	Data requirement	Applicable DID
3.5	Manufacturer's Instruction	*
4.3.3	Report, Test (First Article)	DI-T-2072
4.4.2	Report, Test (Quality Conformance)	DI-T-2072

The DID (Data Item Description Form 1664) for the above data requirements are/will be (*) included in the Department of Defense Authorized Data List (TD-3). Such data will be delivered as identified on completed DIDs when specified on DD Form 1423 (Contract Data Requirements List) and incorporated into applicable contracts.

6.3 Epoxy equivalents. Epoxy equivalents of typical filled systems which have yielded materials conforming to this specification are given in Table II.

Table II. Epoxy Equivalents

Type	Equivalents
I	515-560 g per g mole epoxy
II	850-920 g per g mole epoxy
III	765-830 g per g mole epoxy
IV	515-560 g per g mole epoxy

6.4 First article information. Information regarding submission of first article samples may be obtained from the Naval Undersea Center, (Code 4514), San Diego, CA 92132.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS
Project No. 8030-N057

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-82644(OS), Compound, Epoxy, Encapsulating

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

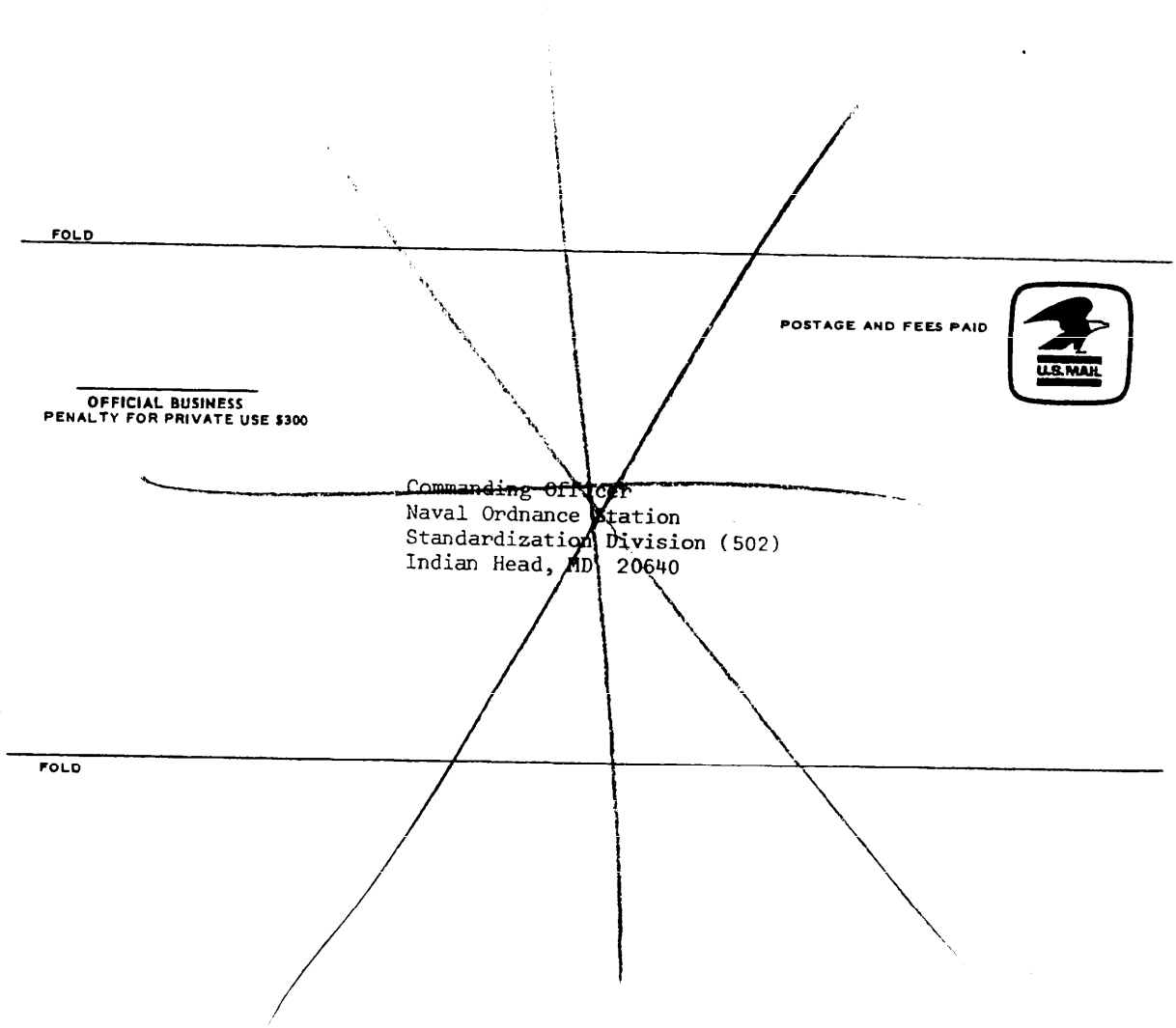
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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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